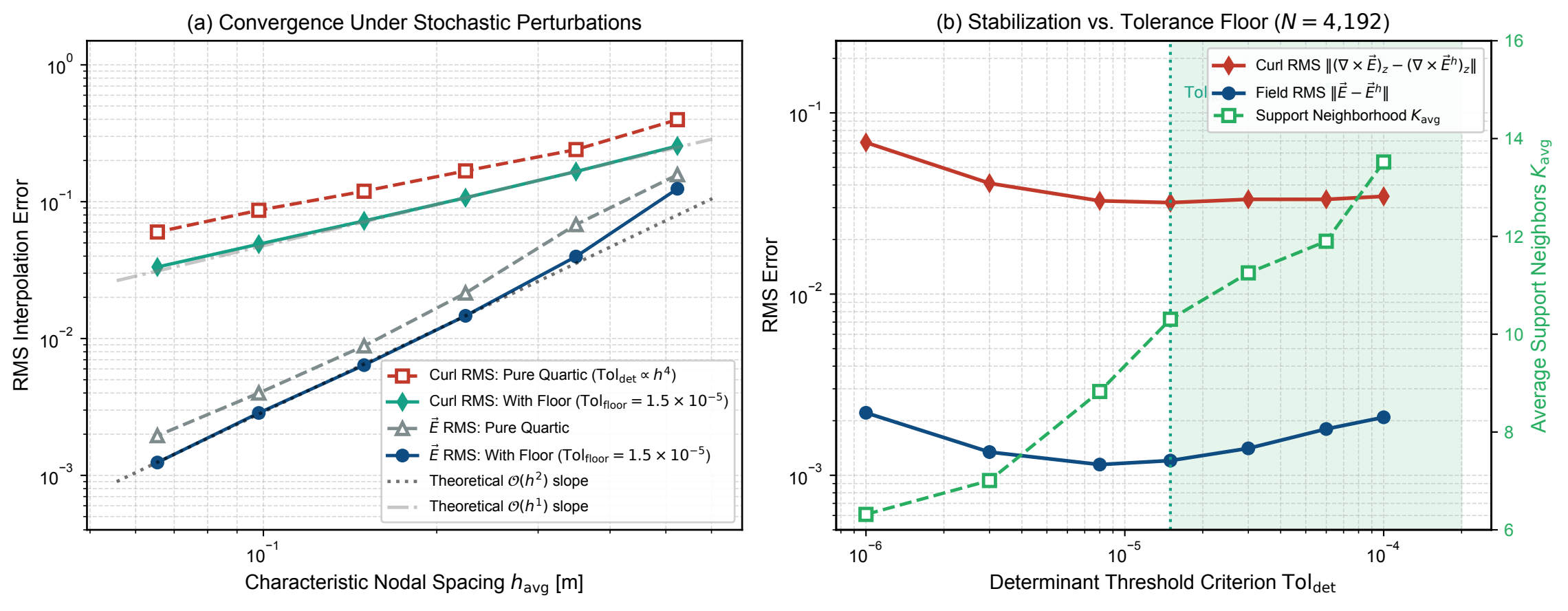
(a) Convergence Under Stochastic Perturbations
RMS Interpolation Error
Characteristic Nodal Spacing h_{avg} [m]
Curl RMS: Pure Quartic ($\text{Tol}_{\text{det}} \propto h^4$)
Curl RMS: With Floor ($\text{Tol}_{\text{floor}} = 1.5 \times 10^{-5}$)
$\vec{E}$ RMS: Pure Quartic
$\vec{E}$ RMS: With Floor ($\text{Tol}_{\text{floor}} = 1.5 \times 10^{-5}$)
Theoretical $\mathcal{O}(h^2)$ slope
Theoretical $\mathcal{O}(h^1)$ slope
(b) Stabilization vs. Tolerance Floor ($N = 4{,}192$)
RMS Error
Average Support Neighbors K_{avg}
Determinant Threshold Criterion Tol_{det}
Curl RMS $\|(\nabla \times \vec{E})_z - (\nabla \times \vec{E}^h)_z\|$
Field RMS $\|\vec{E} - \vec{E}^h\|$
Support Neighborhood K_{avg}